

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of B. Pharm. Examination

University Theory Examination May 2017

PH119/127 Pharmaceutical Chemistry-II

Date: 05.05.2017, Friday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 80

Instructions:

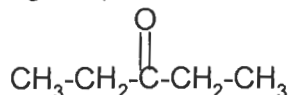
1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat sketches wherever necessary.

Section - I

Q 1 Attempt all questions. Each question is of one mark.

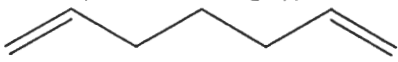
20

1. The order of stability of Carbanion is
[A] Tertiary > Secondary > Primary
[B] Secondary > Tertiary > Primary
[C] Primary > Secondary > Tertiary
[D] Primary > Tertiary > Secondary
2. The IUPAC name of following compound is



- [A] 3-Pentanone
[B] 3-Butanone
[C] 3-Keto Butane
[D] 3-Oxo Pentanone
3. The IUPAC name of isopropyl amine is
[A] 2-Propanamine
[B] 1-Propanamine
[C] 1-Butanamine
[D] Isopropanamine
 4. The SN1 reaction is reaction of alkyl halide with nucleophile in which the product configuration is
[A] Only Walden inversion
[B] Only Retention configuration
[C] Both Walden and Retention of Configuration
[D] None of Above

5. In the molecule of Ethylene, what is the hybridization?
 - [A] sp^3
 - [B] sp^2
 - [C] Unhybrid
 - [D] sp
6. Alkyl halides undergo a type of reaction
 - [A] Nucleophilic substitution
 - [B] Nucleophilic addition
 - [C] Elimination
 - [D] Both [A] & [C]
7. Which of the following has the highest electronegativity
 - [A] S
 - [B] P
 - [C] F
 - [D] O
8. The Saytzeff's rule is
 - [A] It form more stable alkene with greater no. of alkyl group
 - [B] It form more stable alkene with less no. of alkyl group
 - [C] It form least stable alkene with greater no. of alkyl group
 - [D] Both [A] & [C]
9. Which of the following is not an inter molecular force?
 - [A] Repulsion and Attraction
 - [B] Dipole dipole interaction
 - [C] Van der waals forces
 - [D] H-Bonding
10. Which is the example of Homogenous Catalyst?
 - [A] Pd
 - [B] Ni
 - [C] Pt
 - [D] Wilkinson's catalyst
11. Identify the following type's dienes.



 - [A] Conjugated Diene
 - [B] Isolated Diene
 - [C] Cumulated Diene
 - [D] None of above
12. The following statements are correct for carbocation; Except One.
 - [A] Stability order of carbocation is $3^\circ > 2^\circ > 1^\circ$
 - [B] Electron donating group increase the stability of carbocation
 - [C] Carbocation can act as an electrophile
 - [D] Electron withdrawing group increase the stability of carbocation

13. Unit of Dipole Moment is: _____
[A] Cm^{-1}
[B] Poise
[C] Debye
[D] ppm
14. All statement is correct for SN_2 reaction. Except one
[A] Follow Second order kinetic
[B] Rearrangement is possible
[C] Inversion of configuration takes place
[D] Single step reaction
15. Orientation of Elimination reaction follows...
[A] Markoniov's rule
[B] Saytzeff's rule
[C] Micheal addition
[D] Hoffmann rule
16. For which mechanisms the first step involved is the same?
[A] $\text{E}_1 + \text{E}_2$
[B] $\text{E}_2 + \text{SN}_2$
[C] E_1 and SN_1
[D] SN_1 and SN_2
17. Which is not a correct about Ammonia?
[A] SP^3 Hybridization is takes place
[B] Bond angle is 107°
[C] Shape is pyramidal
[D] Bond angle is 105°
18. Which of the following is a polar aprotic solvent?
[A] DMF
[B] Ethanol
[C] Water
[D] Butanol
19. Alkane can prepared by which of following reduction method
[A] Reduction of Alkene
[B] Reduction of Alkyne
[C] Reduction of alkyl halide
[D] All
20. The halogenation of alkane is based on formation of
[A] Carbocation
[B] Carbanion
[C] Free radical
[D] None

SECTION – II

- Q 2** Attempt any Two of the following; **10**
A Describe the stereochemistry of SN_1 and SN_2 reaction 05
B Write Factor Affecting SN_1 and SN_2 reaction 05
C Match the Following; 05

I. Resonance Effect	a. Carbené is an intermediate
II. Huckel's Rule	b. Mesomeric effect
III. Hofmann degradation	c. Inductive effect
IV. Reimer Tiemann reaction	d. Pi Bond
V. Hyper- conjugation	e. Sigma Bond
	f. Nitrene is an intermediate
	g. Carbocation is an intermediate
	h. $4n+2$ π electrons

- Q 3** Attempt any Two of the following; **10**
A Compare SN_1 and SN_2 reaction. 05
B Reason the following statement: 05
 I. 3° Carbocation is more stable than 1° .
 II. SN_2 reaction is faster with 1° alkyl halide.
 III. EDG increase the stability of Carbocation.
 IV. Polar solvent is not suitable for SN_2 reaction.
 V. Triphenyl methyl cation is more stable than Benzyl cation.
C Construct the structure from the following IUPAC names 05
 I. 2,2,3 trimethyl butane
 II. But-1,3-diene
 III. 2-Methyl propanal
 IV. Methanoic acid
 V. Butane dinitrile

SECTION – III

- Q 4** Attempt any FOUR of the following; **20**
A Explain stability of carbocation with examples. 05
B Explain Free radical substitution reaction of alkane. 05
C Write reaction, mechanism, orientation and reactivity of Elimination unimolecular (E_1) reaction. 05
D Write reaction and mechanism for Hoffmann degradation of amide. 05
E Explain 1, 4 addition Vs. 1, 2 addition in dienes. 05
F Write Any three Preparations and reactions of Alkane. 05

Q 5	Attempt any FOUR of the following;	20
A	Write any three preparations and reactions of Alkyl Halide.	05
B	Explain following terms; (Any Five)	05
	I. Electrophile and Nucleophile	
	II. Polar and Non-polar Bond	
	III. Homogenous Catalysis	
	IV. Covalent and Coordinate Covalent Bond	
	V. Inductive Effect	
	VI. Polar Aprotic Solvent	
C	I. Explain Resonance Effect in details.	03
	II. Differentiate SP, SP ² and SP ³ Hybridization.	02
D	Write in detail about the preparation and reaction of Alkynes.	05
E	Write in detail about the preparation and reaction of Alcohol.	05
F	Explain reaction of Alkene with suitable mechanism.	05